Panskura Banamali College (Autonomous)

B.Sc.(H) 1st Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC1T**

Paper Name : Introduction to Programming using C

F.M. : 40 **Time : 2 Hours**

Answer any four questions: **4*10=40**

1. a) Why is C called a mid-level programming language? 2.5 X 4
   b) What are the features of the C language?
   c) What is the difference between the local variable and global variable in C?
   d) What is a Boolean expression?

2. a) What is Macros and Pre processor in C? 2.5 X 4
   b) What is the different between Token and Keyword in C?
   c) What are the Scope rules in C?
   d) Write the difference between Internal and External Linkage in C.

3. a) What is dangling pointer in C? 2.5 X 4
   b) What is Static and Dynamic memory allocation in C?
   c) What is the process to create increment and decrement statement in C? Give an example.
   d) Explain while and while loop with example.

4. a) What are the general description for loop statements and available loop types in C? 2+4+4
   b) What is the correct code to have the following output in C using nested for loop?
   1
   22
   333
   4444
   55555
   c) What is the process(code) to generate random numbers in C programming language?

5. a) What is Typecasting in C? 2.5 X 4
   b) What is the difference between sprintf() and fprintf() in C?
   c) What is Logical and Relational operators, give with example?
   d) How many times the program will print "BCA" , explain?

```
#include<stdio.h>
Int main()
{
printf("BCA");
main();
return 0;
}
```

6. a) What is Recursion in C? 2.5 X 4
   b) Write the difference between Call by value and Call by reference .
   c) Give an example of Multidimensional Array?
   d) What is the limitation of the pointer in C?

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